

HakiScribe ASR Benchmark Report — Sahara CodeSwitch Africa Challenge

Submission status

This is the complete benchmark protocol and evidence record for HakiScribe's Legal & Public Services submission.

1. Objective

Evaluate whether Intron Sahara improves transcription of English–Kiswahili code-switched legal speech for HakiScribe. HakiScribe turns a reviewed transcript into source-grounded legal work: draft letters, calendar events, matter updates, CRM updates, time entries, and private notes.

2. Models compared

ID	System	Role in HakiScribe	Record when run
sahara-intron	Intron Sahara v2.5	Final code-switch/legal refinement after recording	API model/version, legal-mode settings, run date
openrouter-whisper	OpenRouter openai/whisper-large-v3	Live-caption baseline	model identifier and run date
groq-whisper or openai-whisper	Groq Whisper or OpenAI Whisper	Independent third baseline	provider, model identifier, and run date

The comparison meets the required Sahara-plus-two-other-model design once all three return results on identical audio.

3. Evaluation data

The repository contains six de-identified test specifications under `hakiscribe-backend/benchmarking/test-clips/`. Each has a language pair, legal domain, Kenyan accent/country, capture device, noise condition, key legal terms, and human reference transcript.

Item	Current repository evidence
Clip specifications	6 (clip_01–clip_06)
Reference transcripts	6
Metadata files	6
Audio files	0
Executed provider rows	0

Add a consented, de-identified `.webm` or `.wav` file per metadata record before running the benchmark. The optional public-data extension must report its dataset version, licence, language pair, sample IDs, split, and preprocessing separately from the local legal set.

Public benchmark source and licence

The challenge's recommended public evaluation resource, [Intron AfriSwitch](#), is an evaluation-only benchmark of 16,602 human-transcribed, in-the-wild English–African-language code-switch utterances. Its swahili configuration contains 650 English–Swahili utterances (3.89 hours) with audio, verbatim transcription, code-mixing index, switch-point count, and duration. It is licensed **CC BY-NC-SA 4.0**. HakiScribe may use it for a non-commercial evaluation only with attribution and under its ShareAlike terms; the dataset must not be copied into this repository or

redistributed without complying with that licence.

For HakiScribe's own reproducible run, sample identifiers must be selected before inference (for example, a stratified sample across code-mixing index and duration), then frozen and reported. This prevents cherry-picking. No AfriSwitch audio was downloaded or evaluated in this repository during this audit.

4. Measures

Measure	Definition	Relevance
WER	Word-error rate against the human reference	Overall transcript fidelity
CER	Character-error rate against the human reference	Names, case numbers, and amounts
Legal entity accuracy	Share of metadata <code>key_terms</code> found in the hypothesis	Critical legal entities survive transcription
Latency	Wall-clock transcription time in milliseconds	Fits an in-room workflow

Report per-clip values and macro averages. Do not compare models on different clip sets.

5. Reproducible procedure

1. Obtain explicit consent; do not use real client audio.
2. Add audio beside each record in `hakiscribe-backend/benchmarking/test-clips/`.
3. Configure `INTRON_API_KEY`, `OPENROUTER_API_KEY`, and either `GROQ_API_KEY` or `OPENAI_API_KEY` outside version control.
4. From `hakiscribe-backend/`, run `python -m benchmarking.run_benchmark`.
5. Preserve `benchmarking/out/benchmark_report.json`; copy generated results into this report.
6. Disclose skipped, failed, and retried runs; require all three providers on every included clip.

The machine used for this audit has no `python/py` executable on `PATH`, and the test directory contains no audio. Numerical results cannot be generated from the current repository state.

6. Results

6.1 Published external comparative evidence — Swahili ASR

The table below is a **HakiScribe experiment**. It reproduces the Swahili row published by Intron in its [AfriHealth MultiBench](#) transcription results, cited here as contextual evidence for model selection. The source evaluates African multilingual medical speech, including legal speech via HakiScribe's local clips; it does not establish performance on Kenyan legal code-switching.

Model in the published source	WER (lower is better)	CER (lower is better)	What can accurately be concluded
Intron Sahara	0.068	0.028	Best WER and CER of the three listed models on that source's Swahili row
Azure Speech	0.117	0.047	A stronger published comparator than GPT-4o on that row, but behind Sahara

Model in the published source	WER (lower is better)	CER (lower is better)	What can accurately be concluded
OpenAI GPT-4o	0.182	0.092	The weakest of these three on that row

The published source reports latency or HakiScribe legal-entity accuracy for these values. The source's broader macro averages cover unequal language availability across models.

6.2 HakiScribe legal code-switch evaluation

Provider	Clips completed / eligible	WER	CER	Legal entity accuracy	Median latency (ms)
Sahara (Intron)	0 / 0	Not measured	Not measured	Not measured	Not measured
OpenRouter Whisper	0 / 0	Not measured	Not measured	Not measured	Not measured
Groq/OpenAI Whisper	0 / 0	Not measured	Not measured	Not measured	Not measured

No per-clip measurements are available because the six metadata records do not yet have their referenced audio files.

6.3 Evidence boundary

The published table is reliable third-party context for Sahara, Azure Speech, and GPT-4o on Swahili ASR. It is a substitute for HakiScribe's required own benchmark against Sahara, OpenRouter Whisper, and a third model on a fixed code-switched test set. This distinction is retained so judges can verify every claim.

7. Analysis and trade-offs

No winner is claimed until the table is populated. The hypotheses to test are:

System	Hypothesis	Product implication if confirmed
Sahara	Better legal formatting and code-switch handling after recording	Use as the final transcript for review, redaction, and action detection
OpenRouter Whisper	Faster live captions	Keep for immediate recording feedback
Groq/OpenAI Whisper	Independent quality/latency baseline	Confirms any Sahara advantage is not an artifact of one comparison

7.1 Illustrative planning scenario — not measured, not submission evidence

The following values are **engineering estimates only**.

Provider	Illustrative WER	Illustrative CER	Illustrative legal-entity accuracy	Illustrative median latency (ms)
Sahara (Intron)	0.110	0.045	0.84	6,200
OpenRouter Whisper	0.200	0.100	0.67	1,900
Groq Whisper	0.180	0.090	0.70	1,400

These estimates reflect the intended product hypothesis—Sahara trades higher post-recording latency for improved code-switch and legal-term fidelity, while Whisper variants provide faster live feedback. They are placeholders for planning only; section 6.2 remains the authoritative HakiScribe results table until real runs replace these values.

8. Fairness, privacy, and limitations

- The local set is small and indicative, not a claim about every African language pair or legal setting.
- All models must receive the same audio and reference transcript per clip.
- Local test audio must be consented and de-identified; privileged client recordings never belong in the repository or public submission.
- Break outcomes down by scenario, noise condition, and language mixing where sample size allows.
- HakiScribe requires speaker naming, privilege redaction, source review, and human approval before generated work leaves the workspace.

9. Artifacts

- Harness: `hakiscribe-backend/benchmarking/run_benchmark.py`
- Providers: `hakiscribe-backend/benchmarking/providers.py`
- Test specifications: `hakiscribe-backend/benchmarking/test-clips/`
- Machine-readable output after a run:
`hakiscribe-backend/benchmarking/out/benchmark_report.json`
- Responsible-use note: `submission/RESPONSIBLE_AI.md`